

Curriculum Vitae

Wenhao Li, Ph.D. candidate
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EDUCATION

Ph.D.	Southeast University, Nanjing, China <i>Candidate in Transportation Engineering</i> Supervisor: Prof. Yanjie Ji	Sep 2019 – Present
	Nagoya University, Nagoya, Japan <i>Joint Ph.D. Student in Transportation Planning</i> Supervisor: Prof. Tomio Miwa	May 2022 – Nov 2023
M.S.	Northern Arizona University, Flagstaff, AZ, USA <i>Computer Information Technology (Data Analytics)</i>	Jan 2017 – Aug 2019
B.E.	Central South University, Changsha, China <i>Transportation</i>	Sep 2012 – Jun 2016

RESEARCH INTERESTS

Research Domains:

- Intelligent Transportation Systems (ITS):** Traffic Signal Control, Connected & Autonomous Vehicles (CAVs).
- Travel Behavior Analysis:** Route Choice Modeling, Shared Mobility (E-bikes/Bike-sharing), Multi-modal Transport.
- Traffic Flow Theory:** Network Equilibrium, Traffic State Estimation, Trajectory Reconstruction.

Methodologies:

- Data Analytics & AI:** Machine Learning, Deep Learning, Statistical Analysis.
- Simulation & Optimization:** Micro-traffic Simulation (SUMO/VISSIM), Bi-level Optimization, Agent-based Modeling.

RESEARCH EXPERIENCE

Part 1: Academic Research & Grants

- Postgraduate Research & Practice Innovation Program of Jiangsu Province**
 - Project:** *Cooperative Research on Traffic Guidance and Signal Control under Mobile Internet Information* (Grant No. KYCX20_0137)
 - Role:** Project Leader (PI) | Jun 2020 – Jun 2022
 - Key Contribution:** Proposed a collaborative framework for signal control and route guidance using mobile trajectory data; developed bi-level optimization models to improve network efficiency.
- National Key R&D Program of China**

- **Project:** *Synergistic Technology of Urban Traffic Guidance and Signal Control Oriented to Dynamic Network Equilibrium* (Grant No. 2018YFE0120100)
 - **Role:** **Core Researcher** | Jan 2020 – Dec 2021
 - **Key Contribution:** Investigated the interaction between Dynamic Traffic Assignment (DTA) and signal control; designed algorithms to adjust signal phases for network equilibrium.
3. **National Key R&D Program of China**
- **Project:** *Mechanism and Simulation System for Urban Multi-modal Transport Supply and Demand Balance* (Grant No. 2018YFB1600900)
 - **Role:** **Researcher** | 2019 – 2021
 - **Key Contribution:** Built a multi-modal simulation platform to analyze supply-demand characteristics and applied machine learning for peak-hour demand forecasting.

Part 2: Industry & Engineering Projects

1. **Comprehensive Urban Transport System Planning of Hezhou City** (2017 – 2018)
 - Analyzed road network structures to support long-term strategies for public transit and network expansion.
2. **Traffic Analysis & Prediction for Lianshan-Hezhou Highway (Guangxi Section)** (2017 – 2018)
 - Predicted future traffic volume and operating revenue using the Four-Step Model for financial feasibility studies.
3. **Traffic Impact Assessment (TIA) for Guilin Diecai Wanda Plaza** (2018)
 - Simulated traffic flow (VISSIM/Synchro) to evaluate the development's impact and proposed mitigation measures.
4. **Passenger Flow Prediction for Guilin Rail Transit Line 1** (2018 – 2019)
 - Built ridership forecasting models to estimate station-level passenger flow and support station design.

PAPERS (First-Author / *Corresponding-Author)

- 📖 **Li, W.**, Wang, T., & Hu, S. (n.d.). Semantics-guided hierarchical decision-making for autonomous driving via LLM-assisted reinforcement learning. *Transportation Research Part C: Emerging Technologies*. **(Under 2st Review)**
- 📖 Pei, J., Xu, X., & **Li, W***. (n.d.). AVA: Asynchronous vehicular aggregation for efficient distributed training under unstable communication links. *IEEE Transactions on Intelligent Transportation Systems*. **(Under 2st Review)**
- 📖 Wang, T., Wang, B., **Li, W***, & Gu, Z. (n.d.). Extensible hierarchical routing for mixed-autonomy urban networks: MFD-driven region MPC, boundary-flow quotas, and compliance-aware rerouting. *IEEE Transactions on Intelligent Transportation Systems*. **(Under 2st Review)**
- 📖 **Li, W.**, Miwa, T., Li, T., Hu, S., & Wang, T. (n.d.). Dynamic route choice with cognitive updating and information delays: A recursive logit-HMM model on a time-expanded network. *Transportation Research Part E: Logistics and Transportation Review*. **(Under Review)**
- 📖 Pei, J., Bashir, A., Xu, X., & **Li, W***. (n.d.). Agent-assisted adaptive client scheduling for resource-constrained federated learning in AIoT. *IEEE Internet of Things Journal*. **(Under Review)**
- 📖 **Li, W.**, An, Q., & Wang, T. (2026). The Impact of Multisensory Information Processing on Route Change Decisions: An Exploration of Visual and Auditory Effects. *IEEE Intelligent Transportation Systems Magazine*. **(SCI, JCR Q1)**
- 📖 Wang, G., Ma, D., Zhou, L., Alshehri, H. A., & **Li, W.*** (2025). A Secure Multimodal Retrieval Framework for Consumer Electronics under AI-driven Attacks. *IEEE Transactions on Consumer Electronics*. **(SCI, JCR Q1)**
- 📖 **Li W.**, Song J., Ouyang P., et al. (2025). Multi-scale Wavelet-Mamba framework for spatiotemporal traffic forecasting. *Scientific Reports*, 15, 45594. **(SCI, JCR Q1)**

- 📖 **Li, W.**, An, Q., & Ji, Y. (2025). Enhanced navigational insights and their impact on driver route choice: A hybrid utility-regret analysis with heterogeneity. *Research in Transportation Economics*, 110, 101539. (SSCI, JCR Q1)
- 📖 **Li, W.**, Liu, C., Wang, T., & Ji, Y. (2024). An innovative supervised learning structure for trajectory reconstruction of sparse LPR data. *Transportation*, 51(1), 73-97. (SCI, JCR Q1)
- 📖 **Li, W.**, Jia, Y., Ji, Y., Blythe, P., & Li, S. (2024). Modelling autonomous vehicle parking: An agent-based simulation approach. *IET Intelligent Transport Systems*, 18(7), 1237-1258. (SCI, JCR Q2)
- 📖 **Li, W.**, Ji, Y., Cao, X., & Qi, X. (2020). Trip purpose identification of docked bike-sharing from IC card data using a continuous hidden Markov model. *IEEE Access*, 8, 189598-189613. (SCI, JCR Q2)
- 📖 **Li, W.**, Ji, Y., & Wang, T. (2020). Adaptive real-time prediction model for short-term traffic flow uncertainty. *Journal of Transportation Engineering, Part A: Systems*, 146(8), 04020075. (SCI, JCR Q3)
- 📖 **Li, W.**, Yang, Y., Cheng, L., Meng, X., Zhang, F., & Ji, Y. (2023). Understanding adoption intent and behavioral response to shared electric bicycles: A survey in Ningbo, China. *Transportation Research Record*, 2677(1), 1311-1326. (SCI, JCR Q3)
- 📖 An, Q., Qin, Z., Cheng, L., & **Li, W.***. (2025). Assessing the temporal and spatial resilience of bike-sharing demand: A spatiotemporal dynamic analysis. *KSCE Journal of Civil Engineering*, 29(4), 100045. (SCI, JCR Q3)
- 📖 He, X., Sheng, P., Tao, B., & **Li, W.***. (2025). Network Slice-Enabled Federated Learning for Collaborative Traffic Perception in Vehicular Edge Computing. *IEEE Access*, 13, 159109-159126. (SCI, JCR Q2)
- 📖 **Li W.**, Yu J., Qin Z., et al. (2024). Pedal Power vs Battery Boost: An In-depth Analysis of the Dichotomy and Symbiosis between Traditional and Electric Bike-sharing. *Journal of Traffic and Transportation Engineering (English Edition)*, online. (EI)
- 📖 **Li W.**, Ji Y., Wu H., JIA Y., Zhang S. (2024). Empty-load charging strategy for autonomous vehicle parking based on multi-agent system. *Journal of Zhejiang University (Engineering Science)*, 58(8), 1659-1670. (EI)
- 📖 **Li W.**, Ji Y., Qi X., et al. (2021). OD estimation method of urban traffic network based on trajectory reconstruction. *Journal of Southeast University (Natural Science Edition)*, 51(6), 1084-1093. (EI)

Conference Presentations/Posters:

- 📖 **Li W.**, & Ji, Y. (2025). The Impact of Multisensory Information Processing on Lane Change Decisions: An Exploration of Visual and Auditory. *TRB 104th Annual Meeting*, Washington, D.C.
- 📖 **Li W.**, & Ji, Y. (2023). Robust Spatio-Temporal Bike-Sharing Ridership Prediction via Deep Meta Learning. *TRB 102nd Annual Meeting*, Washington, D.C.
- 📖 **Li W.**, & Ji, Y. (2022). A Vehicle Travel Path Reconstruction Method Based on License Plate Recognition Data. *TRB 101st Annual Meeting*, Washington, D.C.

PATENTS

Issued Patents:

- 📖 Ji, Y., Duanmu, H., Liu, P., **Li, W.**, Zheng, Y., & Bu, Q. (2024). *Urban road network OD estimation method based on trajectory reconstruction*. China Patent No. CN202110538868.7.
- 📖 Ji, Y., Yu, J., Liu, P., **Li, W.**, Zhang, F., & Xu, C. (2021). *Signal coordination control method for long arterial intersections based on segmented green wave*. China Patent No. CN112509342B.
- 📖 Ji, Y., Qi, X., **Li, W.**, Liu, P., & Xu, C. (2021). *Vehicle trajectory reconstruction method based on license plate recognition data*. China Patent No. CN112365711B.
- 📖 **Li, W.**, Wan, H., Tian, W., Liang, D., & Feng, R. (2021). *A vehicle detection device for urban traffic intelligent control system*. China Patent No. CN109637155B.

📖 **Li, W., Wang, T., & Li, W.** (2021). *A task switching method, computer device, and readable storage medium*. China Patent No. CN109257432B.

Patent Applications:

📖 **Li, W., Wang, T., & Zhu, W.** (2025). *Road intrusion device for autonomous vehicle testing*. China Patent Application No. CN2025110774670.

📖 **Wang, T., Li, W., & Yang, C.** (2025). *Suspension adjustment device for autonomous vehicles*. China Patent Application No. CN2025109660931.

SOFTWARE COPYRIGHT

📖 **Liu, Y., Li, W., & Li, H.** (2019). *Automated Integration and Processing Software for Vehicle Autonomous Driving Control and Mechanical Equipment V1.0*. China Software Copyright No. 2019SR1337715.

📖 **Li, W., & Wang, T.** (2018). *Urban Traffic Operating Status Identification and Publishing System Based on Parking Data V1.0*. China Software Copyright No. 2018SR900171.

📖 **Li, W., & Wang, T.** (2018). *Vehicle Dynamic Travel Characteristic Analysis System Based on License Plate Information V1.0*. China Software Copyright No. 2018SR617934.

ACADEMIC SERVICES

- **Journal Reviewer:** *Transportation Research Part E; Transportation Research Part D; Journal of Transport Geography; IEEE Transactions on Intelligent Transportation Systems; IEEE ITS Magazine; Transportation Research Record; IET Intelligent Transport Systems; Proc. IMechE Part D: Journal of Automobile Engineering; Multimodal Transportation.*
- **Memberships:** Member, China Society of Automotive Engineers (China-SAE); Student Member, IEEE.

HONORS AND AWARDS

Second Prize, 4th “MathorCup” University Mathematical Modeling Challenge (Big Data Track), 2023

Third Prize, 17th “Huawei Cup” China Post-Graduate Mathematical Contest in Modeling, 2020

Honorable Mention, 9th Mathematical China International Contest in Modeling, 2020

Third Prize, 12th Mathematical China Mathematical Modeling Network Challenge, 2019

Selected Undergraduate Honors:

- Academic Scholarships (Multiple years)
- Awards in Mathematical Modeling Contests (MCM/ICM, APMCM, “Diangong Cup”, etc.)

SKILLS

Technical Skills

- **Programming:** Java, Python, MATLAB, R
- **Traffic Simulation & GIS:** SUMO, TransCAD, ArcGIS
- **Statistical Analysis:** SPSS, Stata

Design & Office Tools

- **Graphic Design:** Adobe Photoshop (PS), Adobe Illustrator (AI)
- **Office Suite:** Microsoft Office (Word, Excel, PowerPoint)

Document & Graphic Design

- **Typesetting & Office:** LaTeX, Microsoft Office (Word, Excel, PowerPoint)
- **Graphic Design:** Adobe Photoshop (PS), Adobe Illustrator (AI)

Certifications

- Associate Constructor (Level II)
- Certified CAD Engineer
- Basketball Referee (Level II)